

# Project

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Based on the game connect 4, how can you develop an MCTS tree search algorithm that can reliably beat a random agent?

- ▶ Create the environment along with the terminal conditions where the reward is 1 for the winner, -1 for the loser and 0 for draw.
- ▶ The opponent is purely random at all times, the play of the opponent can be seen as part of the transition function (no need of the max/min algorithm).
- ▶ You play the blue circles (the opponent has already played first).

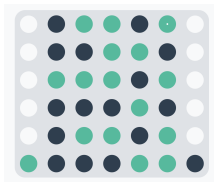


FIGURE – “connect 4” game, blue to play.

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Your need to provide :

- ▶ A report with
  - ▶ a description of the main parts of your algorithm (4 points)
  - ▶ vizualisation of at least one game (with insight about some winning probabilities for the actions) (3 points)
  - ▶ provide meaningful information about the convergence process (3 points)
- ▶ You should provide the source code of your scripts (python). You are allowed to use existing code (not necessarily easier!) as long as you mention the source explicitly both in the code and in the report.
- ▶ Groups of 2
- ▶ Deadline : Dec 8 17 :00.

# Hints for the project

Here are the first things you need to do :

- ▶ A way to encode the state of the game.
- ▶ A way to encode all possible actions given a state.
- ▶ You need to “code” the conditions for a terminal state and whether it is a winning/losing position.